

Technology Stack

Date	23/09/2026
Team ID	SWTID-2026-3089
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Institution	Arignar Anna College Code: MSU 302
Team Members	Aldhaf F (Leader), Bhuvanasri S, Dhanisha Muhi N S, Divya Lakshmi K, Dharshini P
Maximum Marks	2 Marks

Define Your Technology Stack

Identify the platform, tools, workflow engines, database tables, and automation modules used to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with enterprise IT procurement standards.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., Portal, Catalog UI)	ServiceNow Service Portal & Service Catalog (Hardware Category)	Provides an intuitive, responsive self-service interface for employees to request standard laptops with standardized variable forms.
2	Backend / Server-Side (e.g., Process Engine, Logic)	ServiceNow Flow Designer (Process Automation Subsystem)	Low-code/no-code workflow builder that listens to Service Catalog triggers and automatically creates Catalog Tasks assigned to Hardware with elevated System User rights.

3	Database / Data Storage (e.g., Relational Data Tables)	ServiceNow Cloud Relational Datastore (`sc_req_item`, `sc_task`, `alm_hardware`)	Enterprise-grade relational storage maintaining data integrity across requests, task assignments, audit logs, and hardware asset inventory.
4	Cloud / Hosting / Deployment (e.g., SaaS Platform)	ServiceNow Enterprise Cloud Infrastructure (Multi-Tenant SaaS)	High availability (99.9% uptime), automated scaling, SOC 2 / FedRAMP compliance, and zero local hardware maintenance.
5	Version Control & CI/CD (e.g., Artifact Management)	ServiceNow Update Sets & Flow Version Management	Enables version tracking, rollback capability, flow activation states, and structured migration between development, test, and production instances.
6	Third-Party APIs / Other Tools (e.g., Notification & ITIL)	ServiceNow Notification Engine & ITIL v4 Service Management	Manages automated email notifications, approval alerts, task dispatching, and aligns procurement with ITIL best practices.